

Form B

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Title of the Project - JobRix AI-Powered Intelligent Recruitment Platform

Project Summary

JobRix is an AI-powered Intelligent Recruitment Ecosystem that transforms conventional hiring into an automated, skill-centric, and data-driven process. The platform integrates Resume Intelligence, ATS Optimization, Semantic Job Matching, Adaptive AI Interviews, Explainable Candidate Evaluation, and Recruiter Decision Analytics within a unified framework. Unlike existing recruitment portals that provide isolated services, JobRix delivers an end-to-end recruitment lifecycle supported by Artificial Intelligence, Natural Language Processing, Machine Learning, and Large Language Models. The proposed system minimizes recruiter effort, improves candidate-job compatibility, enhances interview quality, and provides continuous career guidance through an AI Career Assistant.

Technical details of the Projects

1. Origin of the Proposal

The current recruitment ecosystem is highly fragmented, where job seekers rely on multiple independent platforms for resume optimization, job discovery, interview preparation, career guidance, and skill assessment, while recruiters use separate Applicant Tracking Systems (ATS), screening platforms, coding assessment tools, and interview software. This fragmented workflow results in increased recruitment time, inconsistent candidate evaluation, poor hiring accuracy, and a lack of personalized career guidance. With the rapid adoption of Artificial Intelligence in Human Resource Technology (HRTech), there is a growing need for an integrated intelligent recruitment ecosystem capable of automating resume analysis, AI-driven job matching, adaptive interview assessment, candidate ranking, recruiter analytics, and personalized career mentoring within a single platform. The proposed system, JobRix, addresses this gap by combining Natural Language Processing (NLP), Machine Learning, Large Language Models (LLMs), RetrievalAugmented

Generation (RAG), semantic similarity search, and AI-based interview evaluation to build a comprehensive recruitment intelligence platform serving both job seekers and recruiters. Unlike conventional recruitment portals that only facilitate job listings or resume uploads, JobRix introduces an end-to-end AI recruitment pipeline capable of assisting users from profile creation to hiring decisions while continuously learning from candidate performance to improve recommendations and hiring outcomes.

2. Definition of the problem

- ✓ Low ATS compatibility causes deserving resumes to be rejected before reaching recruiters.
- ✓ Existing job portals rely primarily on keyword matching rather than semantic skill understanding.
- ✓ Candidates lack realistic interview preparation with personalized feedback.
- ✓ Recruiters spend significant time manually screening thousands of resumes.
- ✓ Existing hiring platforms provide limited insights into candidate strengths, skill gaps, and hiring analytics.
- ✓ Career guidance platforms are disconnected from recruitment workflows.
- ✓ No unified platform currently combines ATS optimization, intelligent job matching, AI interviewing, recruiter analytics, and continuous career mentoring.

3. Objectives

- ✓ **Intelligent Resume Analysis & ATS Scoring** — Parse resumes using NLP to extract skills, experience, education, and certifications, then generate ATS compatibility scores with actionable improvement suggestions.
- ✓ **AI-Powered Job Matching & Recommendations** — Semantically match candidate profiles to job descriptions using AI similarity scoring (beyond keyword matching) to rank and recommend the most relevant opportunities.
- ✓ **Adaptive AI Interview Platform** — Conduct text and voice interviews with dynamically generated questions tailored to the candidate's resume and target role, evaluating technical knowledge, communication, and problem-solving using AI.

- ✓ **Automated Recruiter Workflows & Candidate Screening** — Enable organizations to build AI-driven hiring pipelines that auto-rank applicants by ATS scores and interview performance, reducing manual screening effort significantly.
- ✓ **Career Development & Recruitment Analytics** — Provide personalized career guidance, track skill growth and interview performance, and deliver visual analytics (hiring funnels, skill gap analysis, candidate quality distribution) to support data-driven hiring decisions.
- ✓ **Unified Recruitment Ecosystem** — Integrate all modules — resume analysis, job matching, interviewing, candidate management, and recruiter analytics — into a single scalable platform, eliminating the need for disconnected tools.

4. Work Plan

Methodology

JobRix adopts an **Agile-SDLC integrated with AI/ML techniques** to build an end-to-end intelligent recruitment ecosystem. The development is structured across 8 phases:

1. **Requirement Analysis & System Design** — Study existing platforms (LinkedIn, HireVue, etc.), identify gaps, gather requirements, design three-tier architecture, UI/UX prototypes, and database schema.
2. **Resume Intelligence & Job Matching** — Parse resumes using NLP to extract skills, experience, and certifications; generate ATS scores; perform semantic similarity matching between resumes and job descriptions with explainable "Why You Fit" recommendations.
3. **Adaptive AI Interview Engine** — Dynamically generate interview questions based on resume and job role; support text and live voice modes; evaluate technical knowledge, communication, confidence, and reasoning.
4. **Recruiter Decision Intelligence** — Auto-screen applicants, rank candidates using multi-parameter evaluation, and provide analytics including hiring funnels, skill-gap analysis, and candidate quality insights.
5. **Career Intelligence & Reporting** — Maintain a Career Readiness Profile per candidate, track progress across assessments, and generate personalized AI learning recommendations with downloadable reports.
6. **Testing & Deployment** — Conduct unit, integration, and UAT testing; deploy on cloud infrastructure; continuously monitor and optimize AI models.

Unlike existing platforms, JobRix follows a **closed-loop intelligence pipeline** — Resume → ATS Scoring → Job Matching → AI Interview → Evaluation → Recruiter Analytics → Career Improvement → Re-Assessment.

Modeling and Simulation:

Core AI modules including resume parsing, semantic job matching, interview question generation, and candidate evaluation will be developed and validated using Python-based frameworks (spaCy, HuggingFace Transformers, scikit-learn) in a structured experimental environment. New mathematical and ML models explaining the matching and scoring behavior of the system will be tested against real-world recruitment datasets and benchmarked against each other to investigate the strengths and limitations of each approach, ensuring optimal model selection before deployment.

5. Organization of work element

The work is organized into six key elements.

In the **first element**, the system requirements are gathered by studying existing recruitment platforms such as LinkedIn, Naukri, HireVue, and Indeed, followed by designing the three-tier architecture, database schema, UI/UX prototypes, and API structure for both the Job Seeker and Recruiter portals.

The **second element** involves building the Resume Intelligence Module, where resumes are parsed using NLP techniques to extract skills, education, experience, and certifications, and ATS compatibility scores are generated along with AI-powered improvement suggestions.

In the **third element**, the Intelligent Job Matching Engine is developed, performing semantic similarity matching between candidate profiles and job descriptions to rank and recommend the most relevant opportunities with explainable "Why You Fit" insights.

The **fourth element** focuses on the Adaptive AI Interview Engine, which dynamically generates interview questions based on the candidate's resume and target job role, supports both text and live voice interview modes, and evaluates technical knowledge, communication, confidence, and reasoning ability.

In the **fifth element**, the Recruiter Decision Intelligence module is developed to automatically screen and rank applicants using ATS scores and interview performance, supported by visual analytics including hiring funnels, skill-gap analysis, and candidate quality insights.

Finally, the **sixth element** delivers a unified Career Intelligence system that maintains a Career Readiness Profile for every candidate, tracks improvement across assessments, and generates personalized AI learning recommendations along with downloadable PDF and Excel reports.

6. Time schedule

The work plan can be summarized as follows:

0-3 Months

- Comprehensive literature survey of existing recruitment platforms and AI-based hiring tools.
- Finalization of project scope, research activities, and task scheduling.
- Setting up the development environment and tech stack.
- System design including three-tier architecture, database schema, UI/UX prototypes, and API structure.

3-6 Months

- Development and validation of the Resume Intelligence Module using NLP-based resume parsing, skill extraction, ATS score generation, and improvement suggestions.
- Building the Intelligent Job Matching Engine with semantic similarity matching and explainable "Why You Fit" recommendations.
- Study of different matching parameters such as skill compatibility, experience level, domain relevance, and career preferences affecting recommendation accuracy.

6-9 Months

- Development and testing of the Adaptive AI Interview Engine supporting text and live voice interview modes with dynamically generated questions.
- Implementation of the Recruiter Decision Intelligence module for automated candidate screening, multi-parameter ranking, and visual analytics.
- Identification of key parameters that enhance AI evaluation accuracy and recruiter decision support.

9-12 Months

- Completion of the Career Intelligence module with Career Readiness Profiles, personalized AI learning recommendations, and downloadable reports.
- Validation of AI models and evaluation results against real-world recruitment data and published benchmarks.
- Unit, integration, system, and user acceptance testing followed by cloud deployment.
- Submission of the final project report.

7. Proposed outcome/ findings

The proposed **JobRix** platform is expected to deliver a comprehensive AI-powered recruitment system that automates and simplifies the hiring process for both job seekers and recruiters. The platform will integrate intelligent resume parsing, ATS-based resume evaluation, semantic job matching, AI-driven interview assessment, career recommendations, and recruiter analytics into a single web application. By leveraging Artificial Intelligence, Machine Learning, and Natural Language Processing, the system aims to improve recruitment efficiency, enhance candidate employability, and support data-driven hiring decisions.

The implementation of JobRix is expected to significantly reduce manual resume screening effort, improve candidate-job matching accuracy, and decrease overall recruitment turnaround time. The platform will provide objective candidate evaluation, personalized resume improvement suggestions, intelligent interview feedback, and analytical insights for recruiters, resulting in a more transparent, efficient, and scalable recruitment process. The modular and cloud-ready architecture will also ensure future scalability and seamless integration with emerging recruitment technologies.

Overall, the project is expected to demonstrate the effectiveness of AI in modern recruitment by providing an intelligent, reliable, and user-centric hiring ecosystem. The findings will highlight how AI-based automation can improve hiring quality, minimize manual effort, enhance career guidance, and benefit educational institutions, industries, startups, recruiters, and job seekers through a faster, fairer, and more efficient recruitment framework.

8. System Architecture

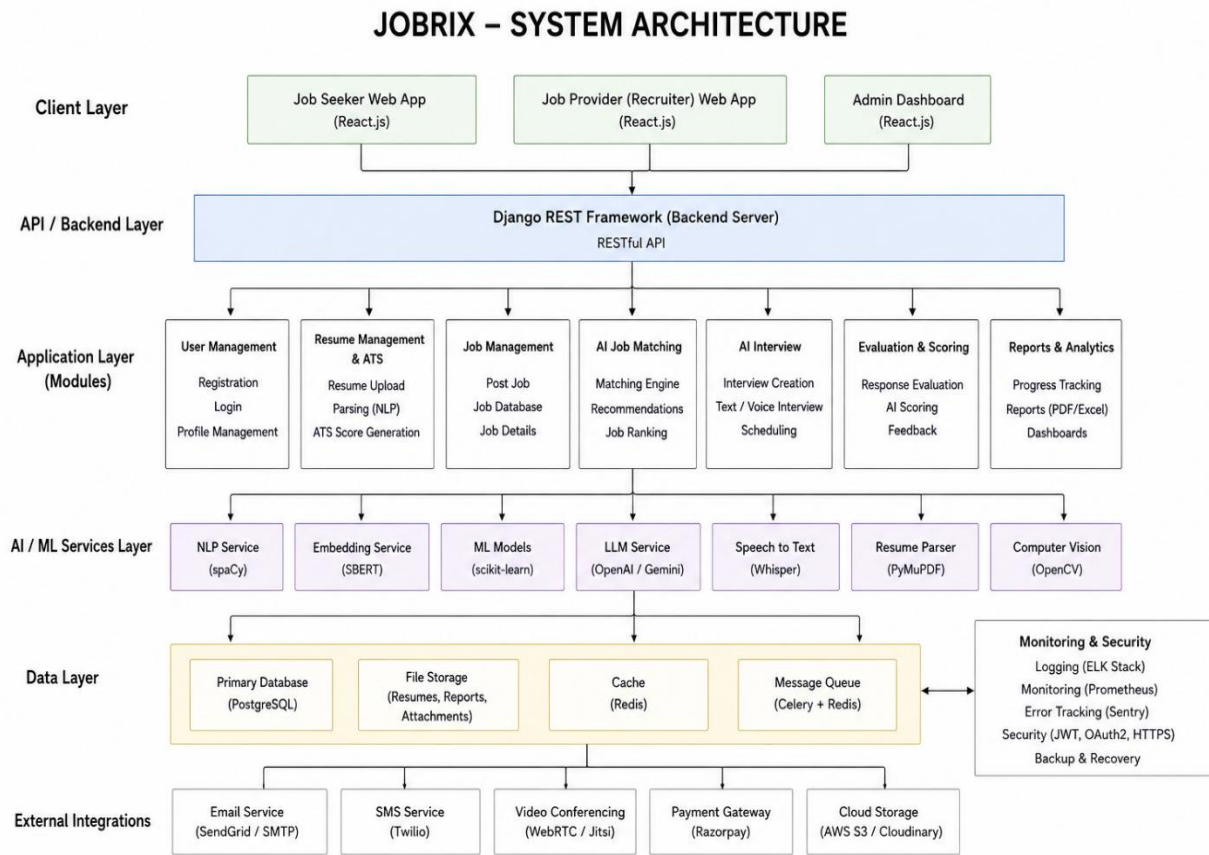


Figure 1

The JobRix platform follows a modular, layered architecture to ensure scalability, security, and maintainability. The client layer consists of separate React.js-based web applications for Job Seekers, Recruiters, and Administrators, which communicate with the backend through RESTful APIs developed using Django REST Framework. The application layer manages core functionalities such as user management, resume parsing and ATS scoring, job management, AI-based job matching, AI interviews, candidate evaluation, and analytics. The AI/ML layer integrates NLP, machine learning, LLMs, speech processing, and resume parsing services to provide intelligent recruitment features. The data layer utilizes PostgreSQL, file storage, Redis cache, and message queues for efficient data management and processing, while monitoring and security services ensure reliable system operation.

9. Budget Estimates (with details):

S. No.	Requirement	Details	Total (₹)
1	Cloud Hosting & Server Infrastructure	Cloud server for application deployment, database hosting, REST API hosting, testing environment, and storage services.	4,000
2	AI API & Model Integration	AI model integration for resume parsing, ATS scoring, semantic job matching, AI interview engine, and Career Assistant (Kai).	3,500
3	Software Development Tools	Development utilities, software libraries, testing tools, debugging tools, version control support, and development environment setup.	2,000
4	Internet & Testing	Internet connectivity for cloud deployment, API testing, system validation, communication, and online development activities.	1,500
5	Documentation & Report Preparation	Printing of project proposal, technical report, user manual, presentation materials, and other documentation.	1,000
6	Contingency & Other Costs	Backup storage, software maintenance, unforeseen development expenses, miscellaneous requirements, and project support activities.	3,000
	Total		₹15,000/-

Table 1

10. Utilization of the Outcome of the Project

The outcomes of the JobRix platform have wide applicability across academia, industry, and research. Educational institutions can integrate the platform into their placement cells to automate resume screening, ATS evaluation, AI-driven interviews, and recruiter analytics, enabling faster,

fairer, and more efficient campus recruitment. Organizations can utilize JobRix as a complete AI-powered recruitment solution to streamline end-to-end hiring, reduce recruitment time and cost, and improve hiring quality through data-driven candidate evaluation.

Job seekers can leverage the platform to evaluate and optimize their resumes, identify skill gaps, practice AI-based mock interviews, receive personalized career guidance, and monitor their career readiness. The AI models developed for resume parsing, semantic job matching, and interview evaluation can further support research in natural language processing, intelligent recruitment systems, and workforce analytics.

Beyond recruitment, JobRix has the potential to democratize access to quality career opportunities by providing AI-powered guidance to candidates from diverse backgrounds, thereby reducing disparities in professional coaching access. Additionally, the recruitment analytics generated by the platform can assist educational institutions, industries, and workforce development agencies in analyzing employment trends, skill demand, and hiring patterns, supporting evidence-based policy formulation and industry-aligned curriculum development.